WORKED SOLUTION

STEP II 1989, Question 16

Probability and Statistics • Binomial Distribution

STEP II 1989, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q16

The key idea

Split the batch into the two parts that behave differently: the n widgets in the sample and the N that are never sampled. The number of faults in each part is binomial, and the two counts are independent. Everything depends on the sample: if it has two or more faults the batch ends up perfect, and otherwise nothing changes. So a batch with $k \geq 1$ faults at the end must have had 0 or 1 faults in the sample, with the rest among the other N.

Solution

Write $q = 1 - p$. Let X be the number of faulty widgets in the sample and Y the number of faulty widgets among the other N in the batch. Each widget is faulty independently with probability p, so

$$X \sim \mathrm{B}(n, p), \qquad Y \sim \mathrm{B}(N, p),$$

and X and Y are independent, because they count faults in disjoint sets of widgets.

Let K be the number of faulty widgets in the batch after quality control. If $X \geq 2$, every fault is corrected and $K = 0$. If $X \leq 1$, nothing is corrected and $K = X + Y$. So

$$K = \begin{cases} X + Y, & X \leq 1, \\ 0, & X \geq 2. \end{cases}$$

The distribution of K

Take $1 \leq k \leq N$. Since $k \geq 1$, the event $K = k$ cannot come from the case $X \geq 2$, so it happens exactly when $X = 0,\ Y = k$ or $X = 1,\ Y = k - 1$. These are disjoint, and by

independence

$$\begin{aligned}\mathrm{P}(K=k) &= \mathrm{P}(X=0)\,\mathrm{P}(Y=k) + \mathrm{P}(X=1)\,\mathrm{P}(Y=k-1)\\ &= q^n\binom{N}{k}p^k q^{N-k} + npq^{n-1}\binom{N}{k-1}p^{k-1}q^{N-k+1}\\ &= \left[\binom{N}{k} + n\binom{N}{k-1}\right]p^k q^{N+n-k}.\end{aligned}$$

Both terms have the same power of p and of q, which is what makes them combine. Now put the binomial coefficients over the common denominator $k!\,(N-k+1)!$:

$$\binom{N}{k} = \frac{N!\,(N-k+1)}{k!\,(N-k+1)!}, \qquad n\binom{N}{k-1} = \frac{n\,N!\,k}{k!\,(N-k+1)!}.$$

Adding, the numerator is $N!\,(N-k+1+nk) = N!\,[N+1+k(n-1)]$, so

$$\mathrm{P}(K=k) = \frac{[N+1+k(n-1)]\,N!}{(N-k+1)!\,k!}\,p^k(1-p)^{N+n-k},$$

as required.

Ask yourself What happens when $k=0$?

The formula only holds for $k \geq 1$. At $k=0$ it gives $\frac{(N+1)N!}{(N+1)!}q^{N+n} = q^{N+n}$, which is the probability that there were no faults at all. But $K=0$ also happens whenever $X \geq 2$, so in fact

$$\mathrm{P}(K=0) = q^{N+n} + \mathrm{P}(X \geq 2) = q^{N+n} + 1 - q^n - npq^{n-1}.$$

The "$k \leq N$" in the question should be read as $1 \leq k \leq N$.

The case $k = N+1$

For $K = N+1$ we need $X \leq 1$ and $X+Y = N+1$. Since $Y \leq N$, this forces $X=1$ and $Y=N$, so

$$\mathrm{P}(K=N+1) = npq^{n-1}\cdot p^N = np^{N+1}q^{n-1}.$$

Putting $k=N+1$ into the formula gives the coefficient

$$\frac{[N+1+(N+1)(n-1)]\,N!}{0!\,(N+1)!} = \frac{(N+1)\,n\,N!}{(N+1)!} = n,$$

and the power of q is $q^{N+n-(N+1)} = q^{n-1}$. So the formula gives $np^{N+1}q^{n-1}$, which is correct. (For $k \geq N+2$ the probability is zero, since $K \leq N+1$.)

The expected number of faults

We could sum $k\,\mathrm{P}(K=k)$, but it is much quicker to use the description of K in terms of X and Y. Writing $\mathbf{1}_{\{X\leq 1\}}$ for the indicator of the event $X \leq 1$,

$$K = (X+Y)\,\mathbf{1}_{\{X\leq 1\}} = X\,\mathbf{1}_{\{X\leq 1\}} + Y\,\mathbf{1}_{\{X\leq 1\}}.$$

For the first term, $X\,\mathbf{1}_{\{X\le 1\}}$ equals 1 when $X = 1$ and 0 otherwise, so its expectation is $\mathrm{P}(X = 1) = npq^{n-1}$. For the second term, Y is independent of X, so

$$\mathrm{E}[Y\,\mathbf{1}_{\{X\le 1\}}] = \mathrm{E}[Y]\,\mathrm{P}(X \le 1) = Np\left(q^n + npq^{n-1}\right).$$

Adding, and taking out the factor pq^{n-1},

$$\begin{aligned}\mathrm{E}[K] &= npq^{n-1} + Np\,q^{n-1}(q + np)\\ &= pq^{n-1}\left[n + N(1-p) + Nnp\right]\\ &= \left[N + n + pN(n-1)\right]p(1-p)^{n-1},\end{aligned}$$

as required.

Checks

- The probabilities should sum to 1. With $n = 2$, $N = 1$: $\mathrm{P}(K = 1) = (2+1)pq^2 = 3pq^2$, $\mathrm{P}(K = 2) = 2p^2q$ and $\mathrm{P}(K = 0) = q^3 + p^2$, and these add to $q^3 + 3pq^2 + 3p^2q + p^3 = 1$ (using $p^2 = p^2(p+q)$).
- If n is huge, almost every batch with a fault in the sample is fixed, and the expectation tends to 0, which matches the factor $(1-p)^{n-1}$.
- We checked every formula with exact fractions for $2 \le n \le 6$, $0 \le N \le 6$ and several values of p, and simulated the procedure: with $n = 4$, $N = 6$, $p = 0.3$ the formula gives $\mathrm{E}[K] \approx 1.585$, and 200 000 simulated batches gave 1.581.

Common slips

- Applying the "two or more" rule to the total number of faults in the batch. Only the sample count X decides whether faults are corrected.
- Forgetting that $K = 0$ has an extra contribution from corrected batches, so the formula fails at $k = 0$.
- Computing $\mathrm{E}[K]$ by brute-force summation. It works, but splitting K with indicators and using independence is far shorter.

Beyond STEP

Writing a random variable as a sum of pieces multiplied by indicators, and using linearity of expectation, is one of the most useful tricks in first-year probability. It avoids finding a distribution when only the mean is needed. The question is also a small model of acceptance sampling in quality control, where we choose n to balance testing cost against the faults that slip through. For more, see Grimmett and Welsh, *Probability: An Introduction*, or Blitzstein and Hwang, *Introduction to Probability*, which uses indicators throughout.

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